# ISA 401: Business Intelligence & Data Visualization

## 01: Introduction to BI and Data Viz

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Automated Scheduler for Office Hours

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# Learning Objectives for Today's Class

- Describe the **analytics journey** and the role of **business intelligence (BI)** within it.
- Differentiate between **structured**, **semi-structured**, and **unstructured** data.
- Map **real-world data sources** to appropriate **data types**.

# ChatISA: Free AI Tools for Every ISA Student

**ChatISA**

**chatisa.fsb.miamioh.edu** · sign in with your Miami account · free for all ISA students

↓

**FOR YOUR COURSEWORK**

- **Coding Tutor** — R & Python help, explained at your level
- **Coding Studio** — Write & run Python, R, and SQL on your data
- **Exam Prep** — Turn course notes into a practice exam
- **Project Assistant** — Plan & pressure-test team projects

**FOR YOUR JOB SEARCH**

- **Job Scout** — This week's analytics & IS jobs, matched to your courses (powers today's dashboard)
- **JobApp Drafter** — Tailor your resume & cover letter to a job
- **Interview Mentor** — Practice a spoken interview for a target job

**GENERAL AI**

- **Ask Anything** — Chat with the frontier model of your choice
- **AI Comparison** — Ask several models at once and compare answers

A **word of caution:** AI answers (ChatGPT, Claude, Gemini, Kimi, ChatISA, etc) can be confidently wrong. The reason you are taking ISA courses is so that **you learn how to verify their outputs and assess their choices**

# An Executive Overview of ChatISA (From HW0)



**ChatISA Executive Overview**
Fadel Megahed






Watch on

# Claude Opus 5 vs GPT-5.6 Sol

03:00

Activity | Class Poll | Notes

- Sign in at https://chatisa.fsb.miamioh.edu.
- Open **AI Comparison** and start a comparison: **Claude Opus 5** vs. **GPT-5.6 Sol**.
- Ask both models this one question:

> *I am taking a business intelligence (BI) course. Give me a short definition of business intelligence. Create a flowchart that captures this process.*

- Read both answers and **pick a winner**.

# The Analytics Journey

# The Analytics Journey: Pre-Analytics [1]

08:00

> *Work with your partner. Explore the data any way you like, including with ChatISA.*

Activity | Your Solution | Discussion

- Skim the source: https://data.cincinnati-oh.gov/Safety/Traffic-Crash-Reports-CPD-/rvmt-pkmq/data
- Download our 2025 extract from Canvas, then answer:

  (a) How many **crashes** are in the dataset?

  (b) How many **variables** does it contain?

  (c) What **dates** does it cover (min and max)?

  (d) Which **day of the week** appears the most in the data? And what does that mean from a **driving-safety** perspective?

# The Analytics Journey: Pre-Analytics [2]

- **Pre-Analytics/Data Management:** where one attempts to **extract** the needed *data* for analysis. Data can either be:



- Stale, uninteresting, convenient
- Highly processed and archived
- Example: `iris`, `mtcars`, `titanic`



- Fresh, interesting, challenging
- Impactful
- Examples: Cincinnati Open Data Portal, Ohio Data Portal, US Government's Open Data.

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While highly processed data can be useful for learning basic concepts, **real-world (often messy) data are much more interesting to work with** — e.g., we can make useful and meaningful decisions from them, as you just did. In this class, we will learn how to scrape, extract, and clean messy data in addition to visualizing clean[ed] data.

**Source:** Slide inspired by Earo Wang's "What I mean by data" (STATS 220, University of Auckland; site no longer online).

# The Analytics Journey: Descriptive [1]

**Descriptive Analytics:** where one attempts to **understand** the data through **descriptive statistics** and **visualizations**.

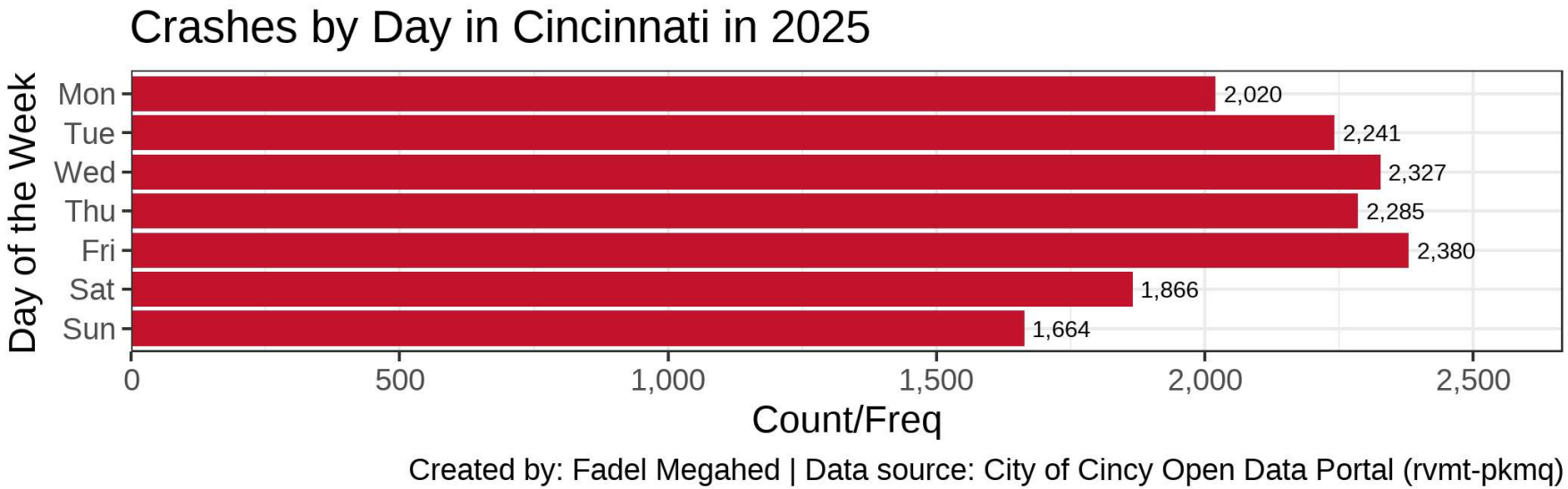
## Descriptive Statistics for 2 Categorical Variables (one row per *crash*)

```
## $dayofweek
##
##  Mon  Tue  Wed  Thu  Fri  Sat  Sun
## 2020 2241 2327 2285 2380 1866 1664
##
## $weather
##
##                    99 - OTHER/UNKNOWN                              4 - RAIN
##                                   178                                 1661
##                             1 - CLEAR                            2 - CLOUDY
##                                 10437                                 1991
##                              6 - SNOW 9 - FREEZING RAIN OR FREEZING DRIZZLE
##                                   476                                    6
##    8 - BLOWING SAND, SOIL, DIRT, SNOW                        5 - SLEET, HAIL
##                                     3                                    9
##                 7 - SEVERE CROSSWINDS                   3 - FOG, SMOG, SMOKE
##                                     2                                   20
```

# The Analytics Journey: Descriptive [2]

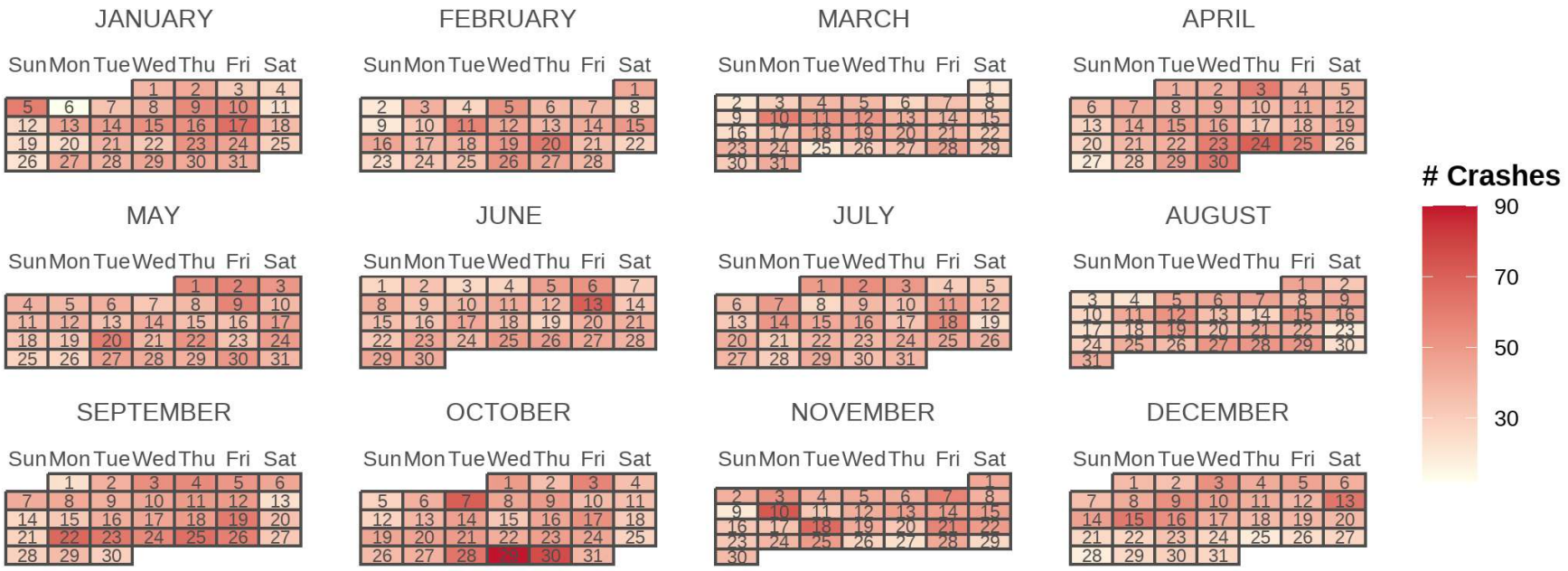
**Descriptive Analytics:** where one attempts to **understand** the data through **descriptive statistics** and **visualizations**.

## A Simple Visualization - A Bar Chart of Crashes Per Day



# The Analytics Journey: Descriptive [3]

**Descriptive Analytics:** where one attempts to **understand** the data through **descriptive statistics** and **visualizations**.




Created by: Fadel Megahed | Data source: City of Cincy Open Data Portal (rvmt-pkmq)

# The Analytics Journey: Continued

## Predictive Analytics

### "What *will* happen?"

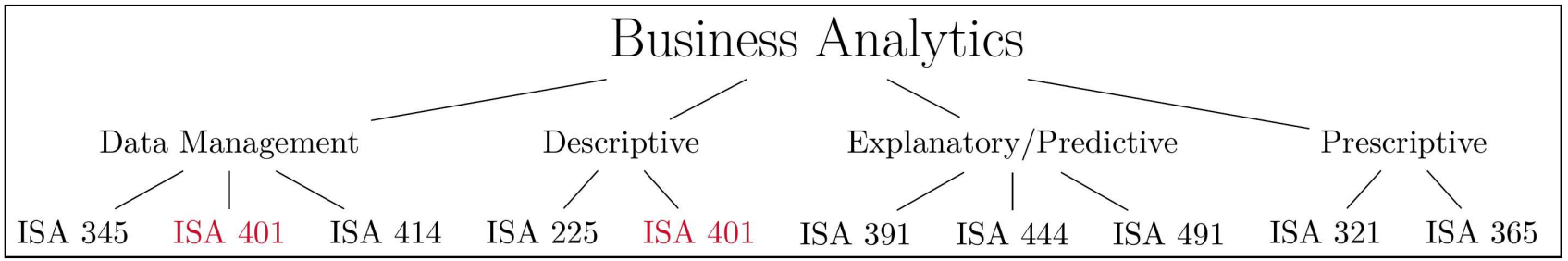
- **Statistical/ML models** trained on historical data.
- Only as good as the data behind them — start simple.

## Prescriptive Analytics

### "What *should* we do?"

- Models that recommend **actions**, not just forecasts.
- e.g., which drivers should a trucking company flag for safety interventions — and at what cost?

# How does our Curriculum at Miami University Prepare you for this Journey?

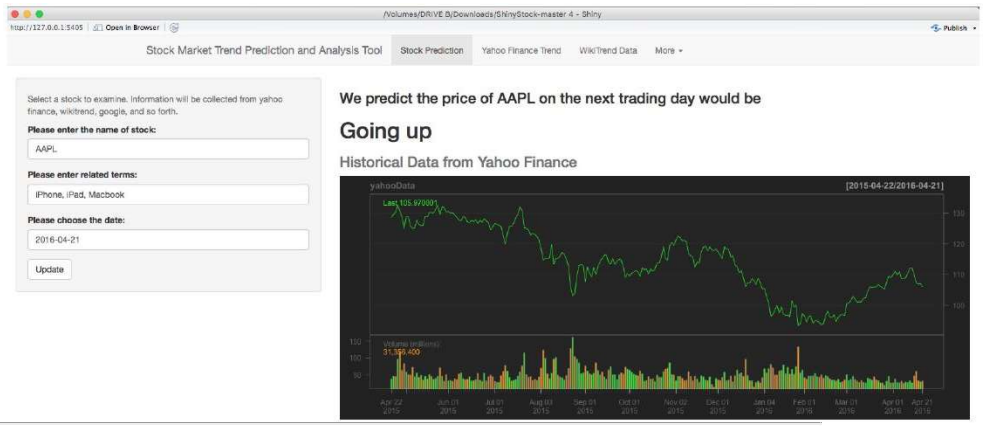



My take on the courses within the business analytics major/minor at Miami University

# Business Intelligence: From Data to Decisions

# What is Business Intelligence?

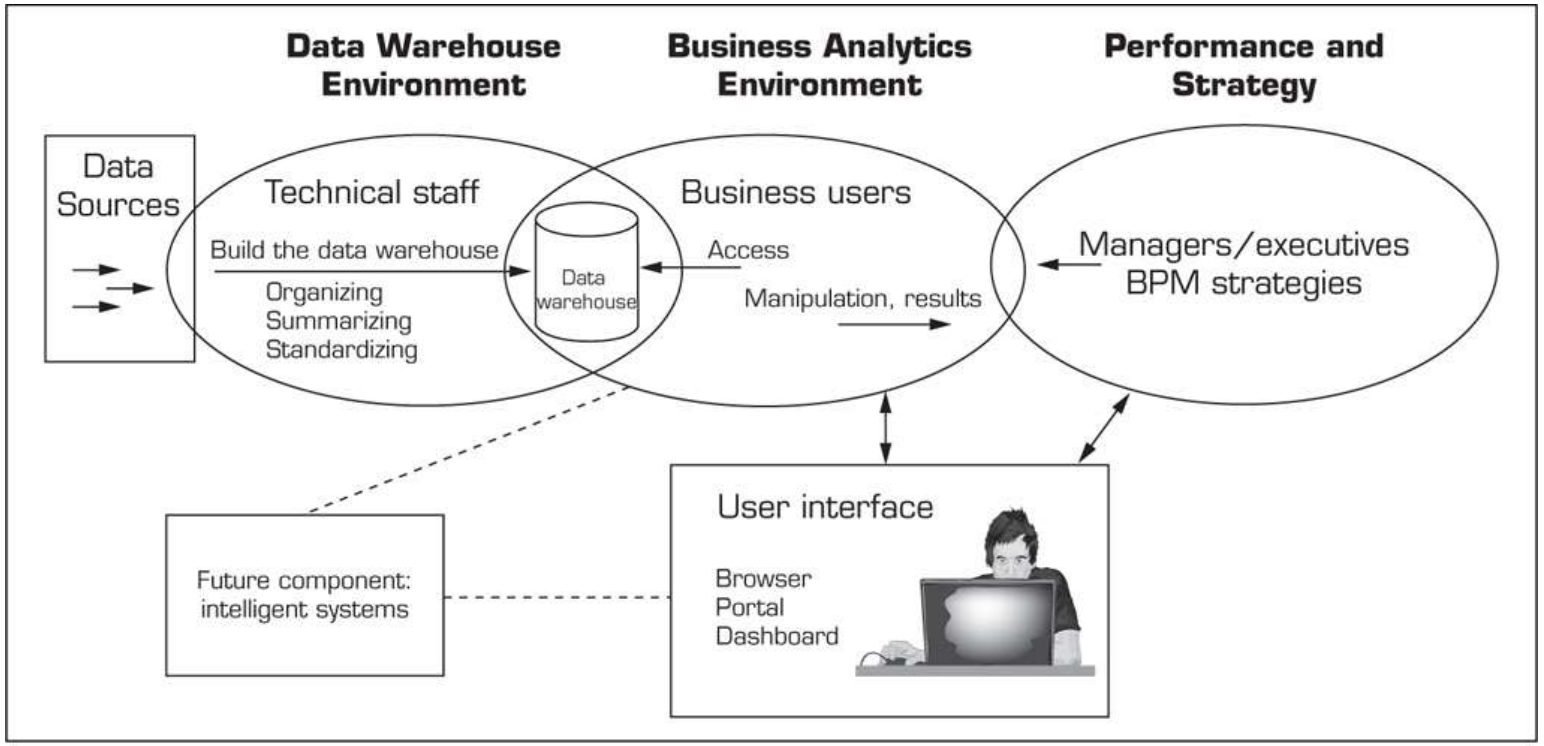
> "... to enable **interactive access (sometimes in real time)** to data, to enable manipulation of data, and to give business managers and analysts the ability to conduct appropriate analysis. By analyzing ... data, situations, and performances, decision makers get valuable insights that enable them to **make more informed and better decisions** ... BI is based on the **transformation of data to information, then to decisions, and finally to actions.**"




**Quote** from Sharda, R., Delen, D., & Turban, E. (2013). Business Intelligence: A managerial perspective on analytics. Prentice Hall Press.

**Image Credit:** Joint work with Bin Weng.

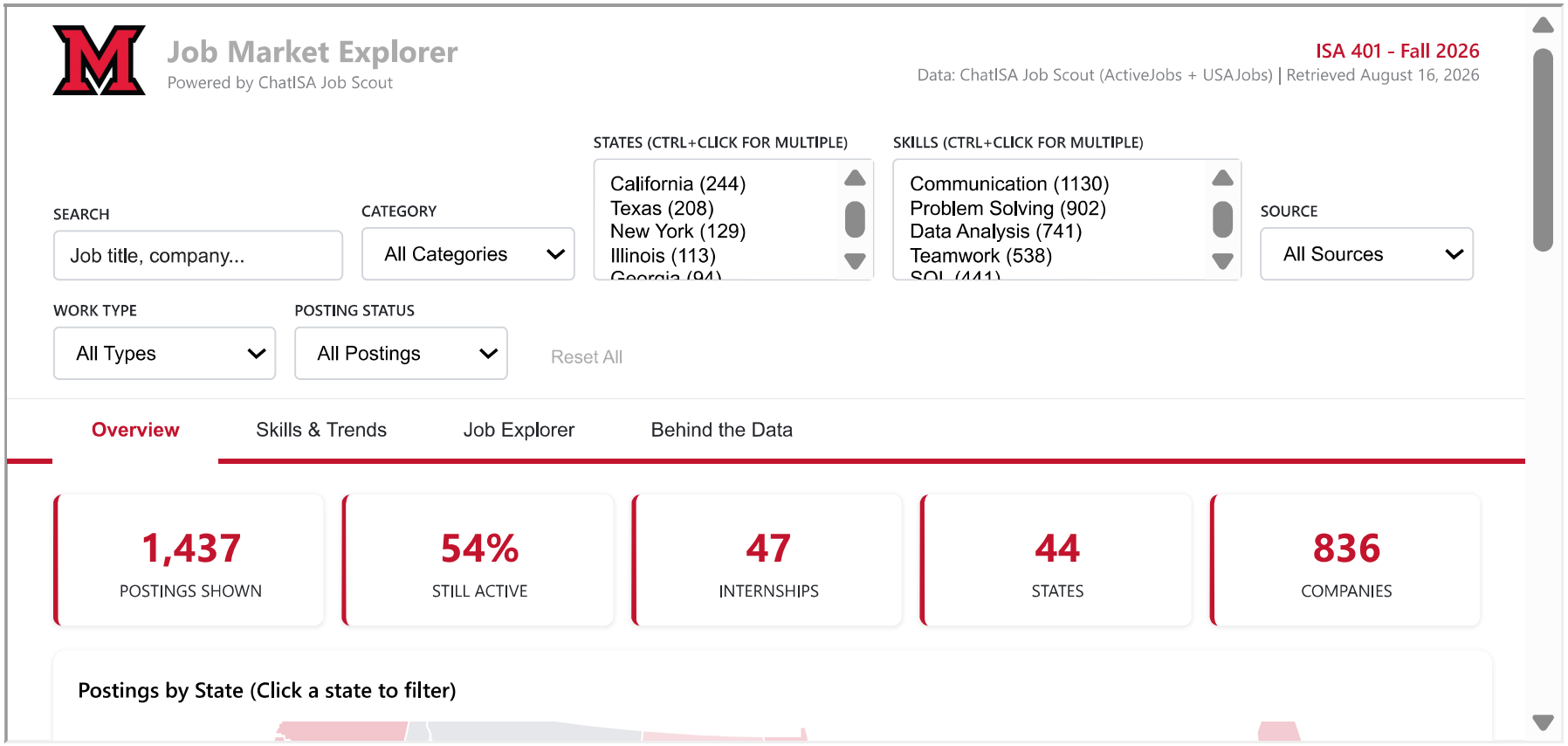
# The BI Process




Source: Based on W. Eckerson, Smart Companies in the 21st Century: The Secrets of Creating Successful Business Intelligent Solutions. The Data Warehousing Institute, Seattle, WA 2003, p. 32, Illustration 5.

**Image Credit:** Sharda, R., Delen, D., & Turban, E. (2013). Business Intelligence: A managerial perspective on analytics. Prentice Hall Press.

# Job Market Explorer: A BI Dashboard in Action




**Data:** 1,437 job postings (full-time, federal, and internships) harvested by **ChatISA Job Scout** from the ActiveJobs and USAJobs APIs. Retrieved August 16, 2026.

# Regarding the Job-Market Dashboard

05:00

Activity | Your Solution | Discussion

Explore the dashboard on your machine (previous slide, or full screen from Canvas). Start in the **Job Explorer** tab and find **one posting you would actually apply to** — internships count! (Notice the *Practice this Interview on ChatISA* button while you are there.) Then answer on Mentimeter. **Write full, specific questions. Think like both a decision-maker and an analyst.**

**Slide 1: Use it**

- What is *one question you would want this dashboard to answer* to help you make a career decision?

**Slide 2: Challenge it**

- What is *one question you would ask about the data itself* before trusting the answer?

# One Job Ad, Three Shapes [1]: Unstructured

UNSTRUCTURED **Research Data Scientist, Magic Eye** at Google (Mountain View, CA; posted 2026-08-12). This is what a human, or Job Scout, actually reads:

MINIMUM QUALIFICATIONS:

* Master's degree in Statistics, Data Science, Mathematics, Physics, Economics, Operations Research, Engineering, or a related quantitative field.
* 3 years of work experience using analytics to solve product or business problems, coding (e.g., Python, R, SQL), querying databases or statistical analysis, or a PhD degree.

PREFERRED QUALIFICATIONS:

* 5 years of work experience using analytics to solve product or business problems, coding (e.g., Python, R, SQL), querying databases or statistical analysis, or a PhD degree.

ABOUT THE JOB:

Magic Eye's mission is to guide strategic decision making for Google's consumer products through data-informed insights, actionable user metrics, and definitions of success. The analysis team partners closely with cross-product area teams from Search, Ads, Android, YouTube, etc., and cross-functionally with teams from Engineering, Behavioral Economics, UX, Finance to bring methodology, data, and insights to Googlers and Google leads. [...] Key research questions include understanding the rapidly evolving GenAI landscape and user journeys across products and devices to identify cross-company strategic opportunities.

Individual pay is determined by factors including job-related skills, experience, and relevant education or training.

US: $147000 - $210000 (USD) + 15% bonus target + equity + benefits

**Unstructured** data is free text with no fixed fields. Notice the **salary is in there** ($147K to $210K): the information exists, but no filter or chart can reach it until someone (or some model) extracts it.

# One Job Ad, Three Shapes [2]: Semi-Structured

SEMI-STRUCTURED Job Scout's AI model reads that text and emits **JSON**: tagged, nested, machine-readable, but with a flexible shape.

```
[
  {"skillId": "python",     "importance": "required"},
  {"skillId": "r",          "importance": "required"},
  {"skillId": "sql",        "importance": "required"},
  {"skillId": "statistical_analysis",
                           "importance": "required"},
  {"skillId": "data_analysis",
                           "importance": "required"},
  {"skillId": "stakeholder_management",
                           "importance": "required"},
  {"skillId": "data_wrangling",
                           "importance": "required"},
  {"skillId": "causal_inference",
                           "importance": "preferred"},
  {"skillId": "experimental_design",
                           "importance": "preferred"},
  {"skillId": "generative_ai",
                           "importance": "preferred"},
```

- **Semi-structured** data carries its own labels (`skillId`, `importance`) but records can differ: this ad has 12 skills, another has 4.
- This is what **APIs** return (JSON like this, or XML/HTML). We'll learn to request and parse it in weeks 4 and 5.
- The *required*/*preferred* labels were decided by an AI model reading the ad: extraction adds structure **and** assumptions.

# One Job Ad, Three Shapes [3]: Structured

STRUCTURED The clean record lands as **one row** in the `scout_postings` table, with the same fields for every posting:

| Field | Value |
|---|---|
| `source` | activejobs |
| `title` | Research Data Scientist, Magic Eye |
| `company` | Google |
| `location_city, location_state` | Mountain View, CA |
| `remote` | 0 |
| `category` | fulltime |
| `posted_at` | 2026-08-12 |
| `active` | 1 |
| `visa_sponsorship` | unknown |

**Structured** data is rows × columns with identical fields throughout. This is the *only* shape the dashboard's filters and charts can use. Notice what didn't survive the trip: there is no `salary` column.

# One Job Ad, Three Shapes [4]: The Big Picture

### UNSTRUCTURED

Free text, images, audio, and video. No fixed fields.

**Our example:** the job-ad text

**Worked with via:** text mining & LLMs (like Job Scout's tagger)

→

### SEMI-STRUCTURED

Self-labeled and nested (JSON, XML, HTML) with a flexible shape.

**Our example:** the skills JSON

**Worked with via:** APIs + parsers (weeks 4–5)

→

### STRUCTURED

Rows × columns with identical fields for every record.

**Our example:** the database row

**Worked with via:** spreadsheets, SQL, R data frames

The closer data gets to **structured**, the less work before analysis. Most interesting data starts on the left.
Much of this course is about moving data from left to right.

# BI Necessitates Merging Data Sources

| Question you asked | Data needed | Possible source | Data type |
|---|---|---|---|
| **Salary** | Wages by occupation | BLS, Glassdoor **(not in our dashboard!)** | STRUCTURED CSV |
| **Skills** | Job requirements | Job-description text, tagged by an LLM | UNSTRUCTURED Text |
| **Location & counts** | Postings by state | ActiveJobs + USAJobs APIs | SEMI-STRUCTURED JSON |
| **Trends** | Weekly harvests over time | Job Scout's SQLite database | STRUCTURED Time series |

**Key insight:** No single source answers everything. You felt this when the dashboard couldn't answer your salary questions. Real analysis requires combining data from multiple places, and that's exactly what we'll learn to do.

# From Data to Decisions: The BI Value Chain

## The Journey

| Stage | Question | Output |
|---|---|---|
| **Data** | What happened? | Raw facts |
| **Information** | What does it mean? | Context + structure |
| **Knowledge** | Why did it happen? | Patterns + insights |
| **Decision** | What should we do? | Action |

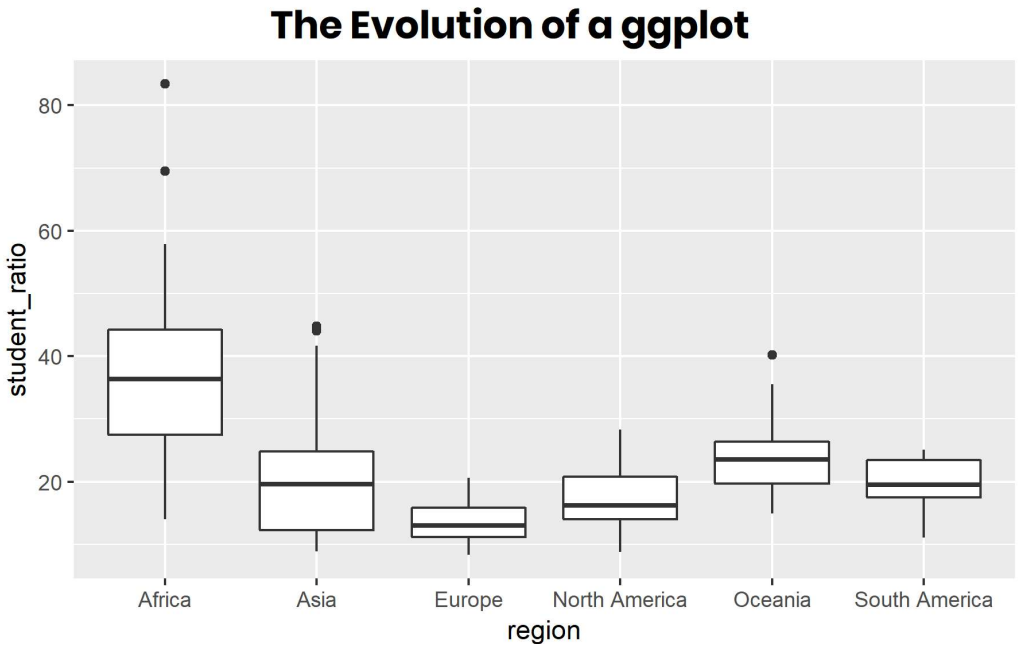
## Example: Job Market

| Stage | Example |
|---|---|
| **Data** | 1,437 postings harvested by Job Scout |
| **Information** | SQL is tagged in 31% of postings; communication in 79% |
| **Knowledge** | Employers want the *pairing*: technical skills **and** the ability to explain findings |
| **Decision** | Build both this semester: code every week, present your results |

# So What is Data Visualization?

# What is Data Visualization?

Data visualization involves **presenting data in a graphical format**. It is really a process that starts by getting data, creating initial plot(s) and modifying them to answer questions of interest. **A good chart takes time and iteration** — for example, see Cedric Scherer's visualization of the UNESCO data on global student to teacher ratios.



# The Goals of Data Visualization

- **Record** information
  - Sometimes a well-designed interactive table *is* the best visualization
- **Analyze** data to support reasoning
  - Develop and assess hypotheses (EDA)
  - Reveal patterns
  - Discover errors in data
- **Communicate** ideas to others
  - Infographics, statistical charts, interactive charts, dashboards
- **Interact with the data (which supports all the above)**
  - You just did this — filtering, mapping, and drilling into the Job Market Explorer

# Analyze Data: The 1854 Cholera Outbreak

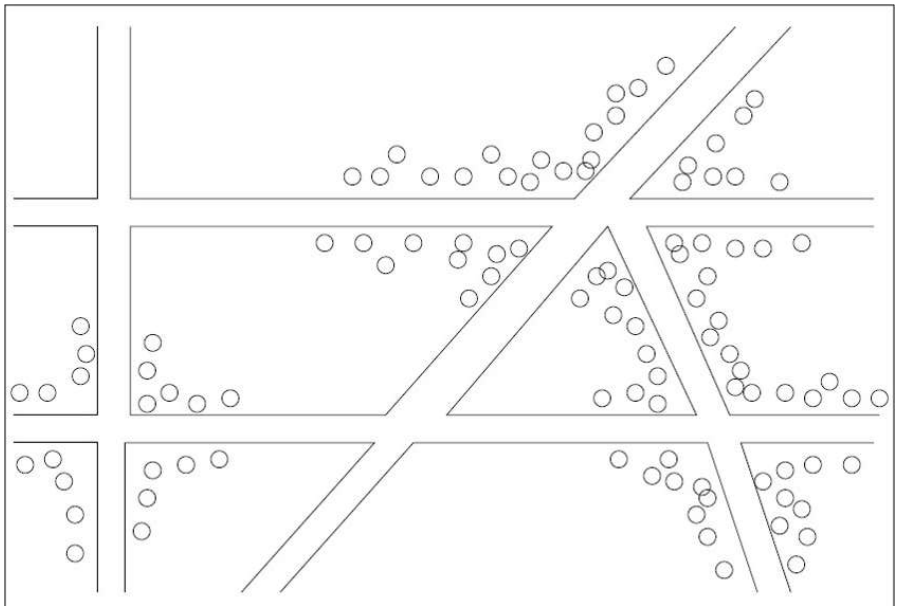


Figure 1.1: Plotting cholera cases on a map of London

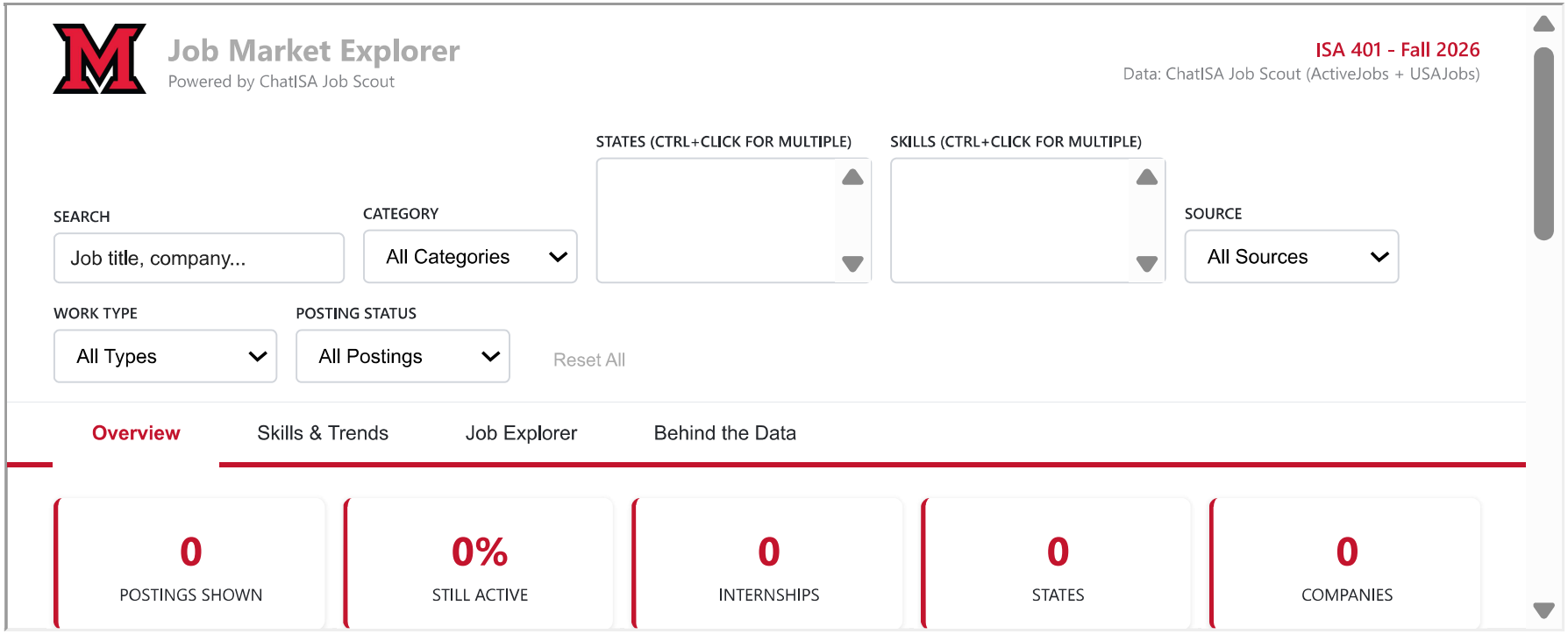
Note the **encoding**: each death is a mark whose **position** on the map carries the information — and we perceive position more accurately than other encoding channels.

Source: Leskovec, J., Rajaraman, A., & Ullman, J. D. (2020). Mining of Massive Data Sets (Third Edition). Cambridge University Press. Image is from Chapter 1, which can be accessed here.

# Critique a Chart from Our Dashboard

05:00

The Chart | Activity | Your Solution

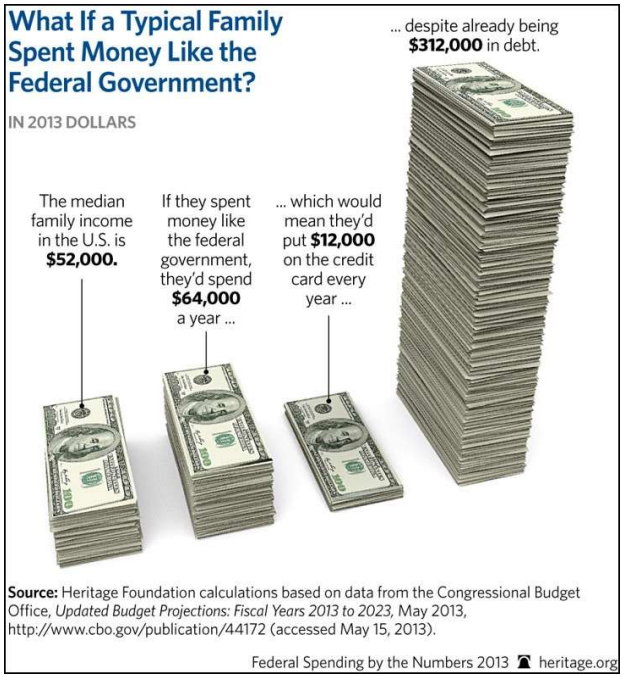


# Communicate Ideas: A Chart in the Wild

04:00

Activity | Your Solution

- What **data** is represented in this visualization? Be specific about the unit of analysis.
- How is the data **visually encoded**? (position, size, color, shape, etc.)
- Is this encoding **effective**? Why or why not?



# Recap

# Summary of Main Points

By now, you should be able to do the following:

- Describe the **analytics journey** (pre-analytics, descriptive, predictive, prescriptive) and the role of **BI** within it.
- Differentiate between **structured**, **semi-structured**, and **unstructured** data — and recognize all three inside one real pipeline.
- Map **real-world data sources** (CSVs, APIs, web pages, free text) to appropriate **data types**.

**Skills you practiced along the way:** identifying the **unit of analysis** (rows ≠ crashes!), distinguishing **counts from rates**, questioning **data provenance**, and reading a chart's **visual encodings**.

# Review and Clarification

- **Class Notes**: Take some time to revisit your class notes for key insights/concepts.
- **Zoom Recording**: The recording of today's class will be made available on Canvas approximately 3-4 hours after the session ends.
- **Optional AI Recaps:** The video and podcast are generated from our slide deck using Gemini Notebook.
- **Questions**: Do not let confusion pile up.

## Slide Deck Video (click to play)




## Slide Deck Podcast (click to play)

# 📖 Prep for Our Next Class 📖

SUGGESTED

### 📈 R for Data Science (R4DS)

**Introduction**
What R and the tidyverse are, and the map of what you'll learn

**Workflow: Basics**
Your first R code: objects, functions, and comments

▶️ Prefer video? **Optional R prep recap on YouTube**

SKIM IF CURIOUS

### 🤖 LLMs

**A Very Gentle Introduction to Large Language Models without the Hype**
How tools like ChatISA actually work, with no math needed

▶️ Prefer video? **Optional LLM recap on YouTube**

**Nothing here is required.** The two R4DS chapters will make our next class (R basics) land much easier. The LLM article is background for the AI tools we will use all semester.

# 🎯 Assignment 🎯

- Complete Assignment 01 on Canvas to reinforce your understanding and application of the topics covered today as well as the suggested prep materials.